

2025 MOCK HSC EXAM

Mathematics Extension 2

General**Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:**100****Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–11)

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Suppose $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$, and $w = \frac{1}{2} - \frac{\sqrt{3}}{2}i$. The value of $\text{Arg}\left(\frac{z^2}{w^2}\right)$ is:

- A. $\frac{7\pi}{6}$ B. $\frac{5\pi}{6}$ C. $-\frac{\pi}{6}$ D. $-\frac{5\pi}{6}$

2 Suppose that f is a continuous, positive function, with domain all real x .

Define $g(x) = \int_{e^{-x}}^{e^x} f(t) dt$. Which of the following is always true about g ?

- A. There exists a real number c such that $g(x) = c$ has no roots.
B. For all real c , there exists exactly one x with $g(x) = c$.
C. g has an inverse on its whole domain, and $g^{-1}(0)$ always exists.
D. For all real c , $g(x) = c$ has multiple solutions.

3 Franz whispers two propositions from the shadowed corridors of his mind into your ear:

A: “If you obey, you endure. If you do not obey, you will not endure.”

B: “If you do not submit, you cannot obey.”

Suppose **A** is known to be true. Which of the following must then be logically equivalent to **B**?

- A. If I endure, then I will submit.
B. If I submit, then I will not obey.
C. If I endure, then I will not obey.
D. If I submit, then I will endure.

- 4 A particle is moving in simple harmonic motion between two points, L and U , with centre O . Given that the particle's speed when it reaches O is $\sqrt{2} \text{ ms}^{-1}$, what is the particle's speed when it is halfway between O and L ?

A. $\frac{\sqrt{3}}{\sqrt{2}} \text{ ms}^{-1}$ B. $\frac{\sqrt{15}}{2} \text{ ms}^{-1}$ C. $\frac{5\sqrt{3}}{3} \text{ ms}^{-1}$ D. $\frac{4\sqrt{3}}{\sqrt{2}} \text{ ms}^{-1}$

- 5 Consider the point $P(2, 2, 3)$, and the line interval ℓ with equation $\vec{r} = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ for $|\lambda| \leq 1$. What is the shortest distance from ℓ to P ?

A. 3 B. $\sqrt{3}$ C. $2\sqrt{2}$ D. $3\sqrt{2}$

- 6 It is known that a polynomial $P(z)$ of degree 13 with real coefficients has a factorisation

$$P(z) = (z^5 + 4)(z - 1 - i)(z^2 - 2z + 2)(z - 1)Q(z),$$

for some polynomial $Q(z)$ with complex coefficients.

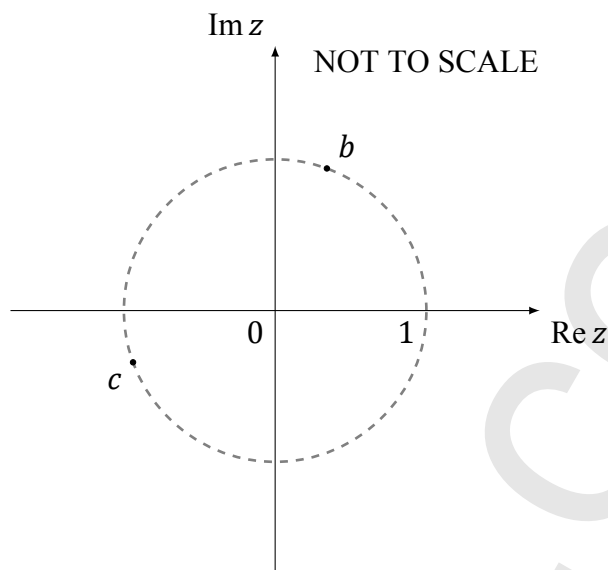
What is the minimum number of real roots (counting multiplicity) $P(z)$ could have?

A. 2 B. 3 C. 4 D. 6

- 7 The value of $\int_0^\pi \frac{x \cos(3x)}{\cos(x)} dx$ is:

A. $\frac{\pi^2}{2}$ B. $\frac{\pi^2 + 1}{2}$ C. $\frac{1 - \pi^2}{2}$ D. $-\frac{\pi^2}{2}$

- 8 Suppose that b and c are complex numbers on the following circle.



It is known that c is a root to the equation $z^2 + az + b = 0$, where a is a complex number. Which of the following is the maximum possible value of $|a|$?

- A. 1 B. $1 - \sqrt{2}$ C. $\sqrt{2}$ D. 2
- 9 There are five tiles on the board, showing
- A, H, 6, 7, BLUE

Each tile has a letter on one side and either a number or a colour on the other (**but not both**).

Suppose that we need to check that following rule is satisfied:

“If a tile shows a vowel, then the other side is an even number or a colour.”

What is the minimum number of tiles needed to check, in order to conclude that the rules have been satisfied?

- A. 1 B. 2 C. 3 D. 4
- 10 Define, for $x \in [1, 87]$, the function $f(x) = \int_1^x \frac{(\ln(t))^2 - 2 \ln(x) + 3}{t} dt$.

Which of the following is closest to the maximum of $f(x)$?

- A. $\frac{14}{3}$ B. $\frac{4}{3}$ C. $\frac{4}{5}$ D. $\frac{16}{5}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

(a) Let $z = 4 + i$ and $w = 2 - i$.

(i) Express $\frac{\bar{z}}{w} - \frac{\bar{w}}{z}$, in the form $x + iy$, where x and y are real numbers. 2

(ii) Express $(z - w)^{12}$ in the form $x + iy$, where x and y are real numbers. 2

(b) Let $f(x) = \frac{1}{(x+1)(2x+3)}$.

(i) Find constants A and B such that $f(x) = \frac{A}{x+1} + \frac{B}{2x+3}$. 1

(ii) Hence, show the volume of the surface of revolution generated by rotating the region under $f(x)$ from $x = 0$ to $x = 1$ is equal to $4\pi \ln \frac{5}{6} + \frac{23\pi}{30} u^3$. 3

(c) Sketch the region defined by $\operatorname{Re}\left(\frac{z-i}{z+i}\right) = 1$ on the Argand diagram. 3

(d) A particle is moving in a straight line such that its displacement from a fixed point, O , at time t is given by $x = 3 - 4 \cos(3t) + 3 \sin(3t)$.

(i) Show that the particle is moving in simple harmonic motion. 1

(ii) What is the maximum velocity of the particle? 1

(iii) When is the displacement of the particle zero for the first time? 2

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) Let a, b , and c be nonnegative real numbers where $a + b + c = 1$. Prove that 1

$$\frac{a}{b^2 + 1} + \frac{b}{c^2 + 1} + \frac{c}{a^2 + 1} \leq 1.$$

- (b) Consider the statement S about a differentiable function f :

$$S: \text{“If } \int_{-1}^1 (1+x)f'(x) dx = 0, \text{ then } f(x) \text{ is not an odd function or } f(1) = 0.”$$

- (i) State the contrapositive of S . 1
- (ii) Hence, prove that S is true. 3
- (iii) Determine whether the converse of S is true. 1
- (c) A vector $\underline{v}$ in 3D satisfies $|\underline{v}|\underline{i} = \underline{v} + 4\underline{a}$, where $\underline{a} = \underline{i} + 2\underline{j} - 2\underline{k}$.
- (i) Find $|\underline{v}|$. 2
- (ii) Find the angle between $\underline{v}$ and $\underline{a}$. 2
- (d) Let x be a six digit number where $a, b, c, d, e, f \in \{1, 2, 3, 4, 5, 6\}$ correspond to the digits of x (so written out, x would read $abcdef$). Note that the digits 0, 7, 8 and 9 are **not** used.
- (i) Suppose x is divisible by 11. Show that the number $a - b + c - d + e - f$ is also divisible by 11. 2
- (ii) Hence or otherwise prove that at least two of the digits of x must be the same. 3

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

(a) Show that none of the roots of $z^6 + z^3 + 1$ lie on the locus of $|z - 1| = |z + 2|$. 2

(b) Let $\mathcal{C}$ be the locus of all possible positions of a point P on the Cartesian plane, given that the distance from P to the point $(-1, 1)$ is equal to the perpendicular distance of P from the line $y = x$.

(i) Let the position vector of P be $\begin{pmatrix} x_p \\ y_p \end{pmatrix}$. Show, using vector methods, that 3

$$(x_p + 1)^2 + (y_p - 1)^2 = \frac{(x_p - y_p)^2}{2}$$

is an equation for $\mathcal{C}$.

(ii) Show that $\mathcal{C}$ is tangent to coordinate axis. 2

(iii) By sketching $\mathcal{C}$, or otherwise, find the area $\mathcal{C}$ encloses with the coordinate axis. 4

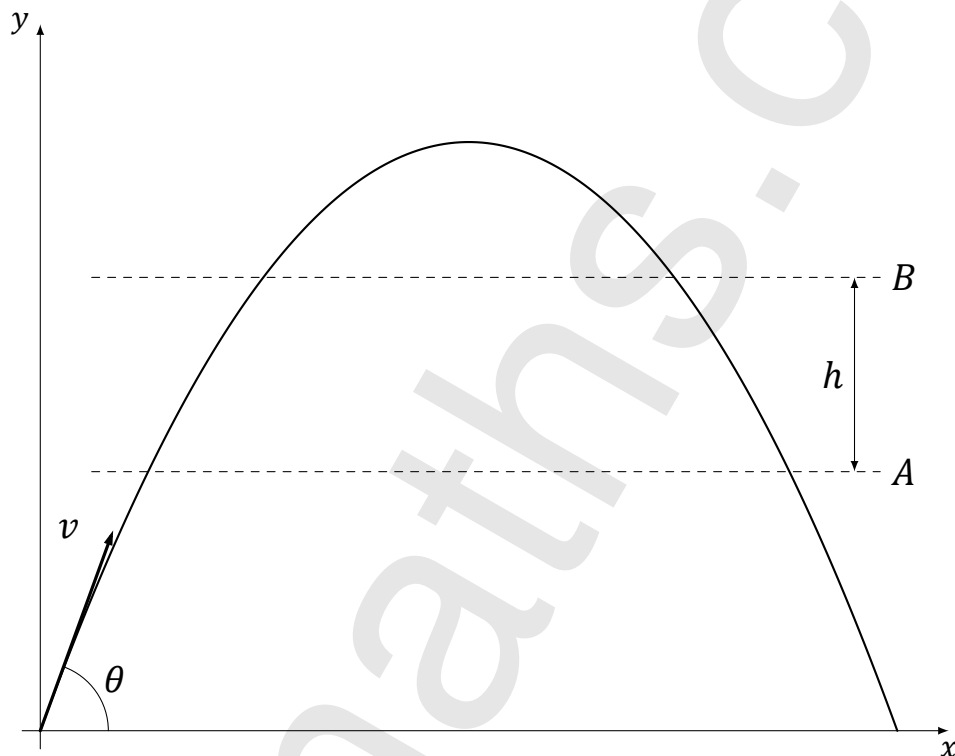
Question 13 continues on page 8

Question 13 (continued)

- (c) A projectile P is projected at an angle θ to the positive x -axis, at an initial velocity v . You may assume that the equation of motion for the projectile is given by

$$\mathbf{r}(t) = \begin{pmatrix} vt \cos \theta \\ vt \sin \theta - \frac{gt^2}{2} \end{pmatrix}, \quad (\text{Do NOT prove this.})$$

where g is the acceleration of gravity.



Lines $y = A$ and $y = B$ have a vertical separation of h . The time between P 's two crossings of lines A and B are T_A and T_B respectively.

- (i) If $\alpha > \beta$ are the real two roots of $bx - ax^2 = c$, find an expression for $\alpha - \beta$. 1
- (ii) Show that $g = \frac{8h}{T_A^2 - T_B^2}$. 3

End of Question 13

Question 14 (16 marks) Use the Question 14 Writing Booklet

- (a) (i) Solve the inequality $\frac{\sqrt{x+1}}{\sqrt{x}-\sqrt{x-1}} > x$ for x . 1

- (ii) Prove by mathematical induction that if n is a positive integer, then 3

$$\frac{\sqrt{2}}{1} + \frac{\sqrt{3}}{2} + \frac{\sqrt{4}}{3} + \cdots + \frac{\sqrt{n+1}}{n} > \sqrt{n}.$$

- (iii) By considering the graph of $y = \frac{\sqrt{x}}{x}$ for $x \geq 1$, prove without mathematical induction that if n is a positive integer, then 3

$$\frac{\sqrt{2}}{1} + \frac{\sqrt{3}}{2} + \frac{\sqrt{4}}{3} + \cdots + \frac{\sqrt{n+1}}{n} > 2\sqrt{n+1} - 2.$$

- (b) An object is initially at rest A units away from a point P . It moves towards P due to gravitational acceleration that is inversely proportional to the square of the distance x of the object from P . Define positive to be away from P .

When the particle is x units away from P , it travels with speed v units towards P .

- (i) Prove that 2

$$v = -\sqrt{\frac{2k(A-x)}{Ax}}$$

where k is some constant of proportionality.

- (ii) Let T be the number of seconds it takes for the object to reach the half-way point between its initial position and P . 4

Show that $T = \frac{A\sqrt{A}(\pi + 2)}{4\sqrt{2k}}$.

- (c) Using the substitution $u = x\sqrt{x}$, evaluate $\int_0^1 \sqrt{\frac{3x}{3+x^3}} dx$. 3

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

(a) Suppose $n \in \mathbb{Z}^-$. Let $I_n = \int_0^{\frac{\pi}{2}} (\sin x + 1)^n dx$.

(i) Show that

$$I_n = I_{n+1} - 1 - n \int_0^{\frac{\pi}{2}} \cos^2 x (1 + \sin x)^{n-1} dx.$$

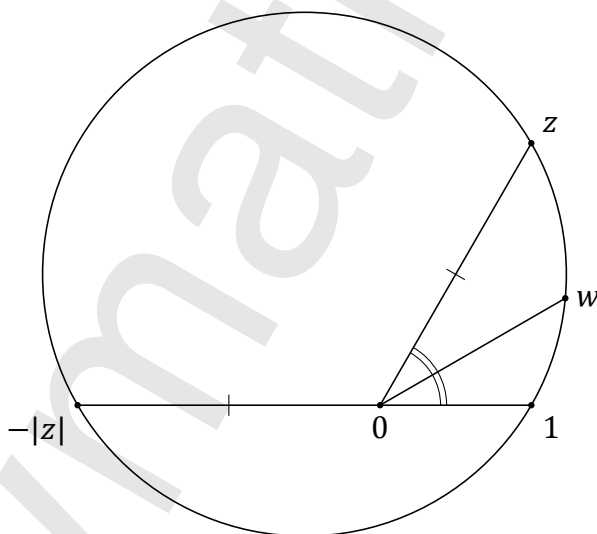
(ii) Hence, show that

$$I_n = \frac{n+1}{2n+1} I_{n+1} - \frac{1}{2n+1}.$$

(iii) Show that $\int_0^{\frac{\pi}{2}} \frac{dx}{(1 + \sin x)^4} = \frac{12}{35}$.

(b) Let z and w be two complex numbers in the first quadrant of the Argand diagram, with $\text{Arg}(z) = 2 \text{Arg}(w)$. It is known that there exists a circle which passes through the four complex numbers $1, z, w$, and $-|z|$.

This information is shown in the following diagram.



(i) Explain why there is a positive real number k satisfying $kw^2 = z$.

(ii) By considering two equal angles subtended by the same arc, prove that

$$\frac{(z-1)(w+|z|)}{(w-1)(z+|z|)}$$

is purely real.

(iii) Deduce that w is one of the square roots of z .

Question 16 (14 marks) Use the Question 16 Writing Booklet

(a) Suppose $\underline{a}$, $\underline{b}$, and $\underline{c}$ are nonzero 3D vectors satisfying the following three conditions:

1. $|\underline{a} + \underline{b}| = |\underline{a} - \underline{b}| = |\underline{c}|$,

2. $|\underline{a}| = 3|\underline{b}|$, and

3. $(2\underline{a} + \underline{b})$ is parallel to $\underline{c}$.

(i) Show that $\underline{a} \cdot \underline{b} = 0$.

1

(ii) Explain why there exists a real number λ satisfying $2\underline{a} + \underline{b} = \lambda\underline{c}$, and show further that λ satisfies

2

$$\lambda^2 = \frac{4|\underline{a}|^2 + |\underline{b}|^2}{|\underline{a}|^2 + |\underline{b}|^2}.$$

(iii) Find all possible values of $\underline{c}$ in terms of $\underline{a}$ and $\underline{b}$.

2

(b) Suppose $n \geq 2$ is a positive integer, and let $\omega = e^{\frac{\pi i}{n}}$ be an $2n$ th root of unity. Consider the polynomial $P(z) = z^{2n} - 1$.

(i) Show that $2n = |(1 - \omega)(1 - \omega^2) \cdots (1 - \omega^{2n-1})|$.

2

(ii) Hence, show that $\sum_{k=1}^{n-1} \ln \left(\sin \frac{\pi k}{2n} \right) = \frac{1}{2} \ln n - (n-1) \ln 2$.

4

(c) Consider the function $f(x) = \ln(\csc x)$ for $x \in \left(0, \frac{\pi}{2}\right]$.

3

You may assume that $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$ (Do NOT prove this).

Find an expression for S_n , the sum of the areas of n lower rectangles for $f(x)$ over the interval $\left(0, \frac{\pi}{2}\right]$, and hence conclude using (b) that

$$\int_0^{\frac{\pi}{2}} \ln(\csc x) dx = \frac{\pi}{2} \ln 2.$$

End of paper

*To look back one day, through all the sorrow,
 the celebration, the desperation, the 3AM pleas on ears that were tired of listening,
 See the paths I didn't take,
 But appreciate the one I did choose.
 To tell people I lost: it was quite fortunate to have met you.
 To tell myself that the clear blue night would come.
 To find forgiveness for carrying the hopes and dreams of the dead,
 for they would want you to carry your own for the future.*

*Look forward to the once obscured starlight,
 the shadows it casts dimming with every step towards sunrise.*

EXAMINERS

neutde	Q2, 4, 5
one eight seven	Q11[a,d], Q12[b,c], Q13a
Hugh Entwistle	Q9, Q12d, Q14b
MagiicMushroom	Everything else.

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

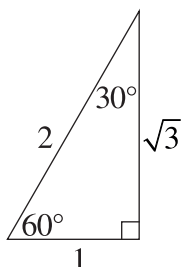
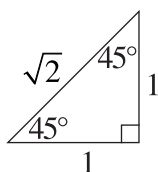
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

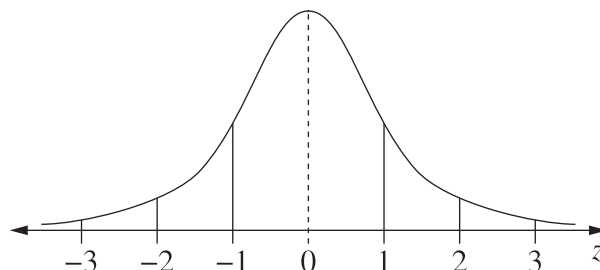
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$